

Week 7&8_Hembree_A

July 5, 2026

0.0.1 DSC 540 - Data Preparation

0.0.2 Week 7 & 8 Exercises

The four chapters you read these two weeks focus extensively on cleaning and transforming data.

You can choose from either of these two datasets:

So Much Candy Data, Seriously. (Ng, 2017) - The Metropolitan Museum of Art Open Access CSV. (Github, 2019 (this data set has multiple years' worth of data – you can use these files for merging/joining) - You can also download all of the above data from both sites directly from this link: Weeks 7 & 8 Datasets

For this assignment you need to complete 8 of the following exercises against this data.

Note: You must select at least two methods from each chapter to perform on one of the datasets. You are welcome to do more methods and you do not have to use the same dataset for all 8 methods.

Chapter 7 - Filter out missing data - Fill in missing data - Remove duplicates - Transform data using either mapping or a function - Replace values - Discretization and Binning - Manipulate Strings

Chapter 8 - Create hierarchical index - Combine and Merge Datasets (you will have to either create a new dataset from your existing data or create a relationship between the data I have provided) - Reshape - Pivot the data

Chapter 10 - Grouping with Dicts/Series - Grouping with Functions - Grouping with Index Levels - Split/Apply/Combine - Cross Tabs

Chapter 11 - Convert between string and date time - Generate date range - Frequencies and date offsets - Convert timestamps to periods and back - Period Frequency conversions

```
[1]: # Import Libraries
import pandas as pd
import numpy as np

# Read in file
metobj_df = pd.read_csv("C:/Users/Ashem/OneDrive/Desktop/Masters Program - DSC/
↳DSC 540 - Data Preparation/Week 7&8 Data/MetObjects.csv")

# Preview file
metobj_df.head()
```

C:\Users\Ashem\AppData\Local\Temp\ipykernel_36936\3196131170.py:6: DtypeWarning: Columns (5,7,10,11,12,13,14,34,35,36,37,38,39,40,41,42,43,44,45,46) have mixed types. Specify dtype option on import or set low_memory=False.

metobj_df = pd.read_csv("C:/Users/Ashem/OneDrive/Desktop/Masters Program - DSC/DSC 540 - Data Preparation/Week 7&8 Data/MetObjects.csv")

```
[1]:  Object Number  Is Highlight  Is Timeline Work  Is Public Domain  Object ID  \
0      1979.486.1          False          False          False          1
1      1980.264.5          False          False          False          2
2        67.265.9          False          False          False          3
3        67.265.10         False          False          False          4
4        67.265.11         False          False          False          5
```

```
Gallery Number      Department AccessionYear Object Name  \
0              NaN  The American Wing      1979.0      Coin
1              NaN  The American Wing      1980.0      Coin
2              NaN  The American Wing      1967.0      Coin
3              NaN  The American Wing      1967.0      Coin
4              NaN  The American Wing      1967.0      Coin
```

```
                Title ... River Classification  \
0  One-dollar Liberty Head Coin ...      NaN      NaN
1  Ten-dollar Liberty Head Coin ...      NaN      NaN
2   Two-and-a-Half Dollar Coin ...      NaN      NaN
3   Two-and-a-Half Dollar Coin ...      NaN      NaN
4   Two-and-a-Half Dollar Coin ...      NaN      NaN
```

```
Rights and Reproduction                                Link Resource  \
0              NaN  http://www.metmuseum.org/art/collection/search/1
1              NaN  http://www.metmuseum.org/art/collection/search/2
2              NaN  http://www.metmuseum.org/art/collection/search/3
3              NaN  http://www.metmuseum.org/art/collection/search/4
4              NaN  http://www.metmuseum.org/art/collection/search/5
```

```
Object Wikidata URL Metadata Date                                Repository  \
0              NaN      NaN  Metropolitan Museum of Art, New York, NY
1              NaN      NaN  Metropolitan Museum of Art, New York, NY
2              NaN      NaN  Metropolitan Museum of Art, New York, NY
3              NaN      NaN  Metropolitan Museum of Art, New York, NY
4              NaN      NaN  Metropolitan Museum of Art, New York, NY
```

```
Tags Tags AAT URL Tags Wikidata URL
0  NaN      NaN      NaN
1  NaN      NaN      NaN
2  NaN      NaN      NaN
3  NaN      NaN      NaN
4  NaN      NaN      NaN
```

[5 rows x 54 columns]

Ch 7 transformations

- Filter out missing data
- bin by Accession Year (not actually, went with manipulating the strings instead for reasons later explained)

```
[2]: # Check out the data
metobj_df.info
```

```
[2]: <bound method DataFrame.info of          Object Number  Is Highlight  Is Timeline
Work  Is Public Domain  \
0      1979.486.1      False          False          False
1      1980.264.5      False          False          False
2       67.265.9      False          False          False
3       67.265.10     False          False          False
4       67.265.11     False          False          False
...      ...      ...      ...      ...
484951  55.621.134     False          False          False
484952  1977.646     False          False          False
484953   33.40.1     False          False          False
484954  170.1 C42      True          False          False
484955  17.3.3457     False          False          False
```

```
          Object ID Gallery Number      Department AccessionYear  \
0              1      NaN  The American Wing      1979.0
1              2      NaN  The American Wing      1980.0
2              3      NaN  The American Wing      1967.0
3              4      NaN  The American Wing      1967.0
4              5      NaN  The American Wing      1967.0
...      ...      ...      ...      ...
484951      900605      NaN  Drawings and Prints      1955
484952      900606      NaN  Drawings and Prints      1977
484953      900633      NaN  Drawings and Prints      1933
484954      900717      NaN      The Libraries      NaN
484955      900748      NaN  Drawings and Prints      1917
```

```
          Object Name      Title ...  \
0              Coin  One-dollar Liberty Head Coin ...
1              Coin  Ten-dollar Liberty Head Coin ...
2              Coin  Two-and-a-Half Dollar Coin ...
3              Coin  Two-and-a-Half Dollar Coin ...
4              Coin  Two-and-a-Half Dollar Coin ...
...      ...      ...      ...
484951      Print      Holiday Card ...
484952      Print      Brooklyn Local ...
```

484953	Print	Yesterday and Today ...
484954	NaN	De la loi du contraste simultan� des couleurs,... ...
484955	Print	Madison Square - Fifth Avenue and Broadway at

	River	Classification	Rights and	Reproduction \
0	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN
...	...	...	...	...
484951	NaN	Prints	NaN	NaN
484952	NaN	Prints	NaN	NaN
484953	NaN	Prints	NaN	NaN
484954	NaN	NaN	NaN	NaN
484955	NaN	Prints	NaN	NaN

	Link	Resource	Object	Wikidata	URL \
0	http://www.metmuseum.org/art/collection/search/1			NaN	
1	http://www.metmuseum.org/art/collection/search/2			NaN	
2	http://www.metmuseum.org/art/collection/search/3			NaN	
3	http://www.metmuseum.org/art/collection/search/4			NaN	
4	http://www.metmuseum.org/art/collection/search/5			NaN	
...	...	...	...	...	...
484951	http://www.metmuseum.org/art/collection/search...			NaN	
484952	http://www.metmuseum.org/art/collection/search...			NaN	
484953	http://www.metmuseum.org/art/collection/search...			NaN	
484954	http://www.metmuseum.org/art/collection/search...			NaN	
484955	http://www.metmuseum.org/art/collection/search...			NaN	

	Metadata	Date	Repository	Tags \
0	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN
1	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN
2	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN
3	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN
4	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN
...	...	...	...	...
484951	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN
484952	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN
484953	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN
484954	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN
484955	NaN	Metropolitan Museum of Art, New York, NY	NaN	NaN

	Tags	AAT	URL	Tags	Wikidata	URL
0	NaN			NaN		
1	NaN			NaN		
2	NaN			NaN		

3	NaN	NaN
4	NaN	NaN
...	...	...
484951	NaN	NaN
484952	NaN	NaN
484953	NaN	NaN
484954	NaN	NaN
484955	NaN	NaN

[484956 rows x 54 columns]>

```
[3]: # Determine how many na values exist
metobj_df.isna().sum()
```

```
[3]: Object Number          0
      Is Highlight          0
      Is Timeline Work      0
      Is Public Domain      0
      Object ID             0
      Gallery Number        435415
      Department            0
      AccessionYear         3862
      Object Name           2266
      Title                 28803
      Culture               276766
      Period                393813
      Dynasty               461755
      Reign                 473720
      Portfolio             458442
      Constituent ID        202443
      Artist Role           202443
      Artist Prefix         202443
      Artist Display Name   202443
      Artist Display Bio    204533
      Artist Suffix         202491
      Artist Alpha Sort     202443
      Artist Nationality    202443
      Artist Begin Date     202443
      Artist End Date       202443
      Artist Gender         378474
      Artist ULAN URL       257515
      Artist Wikidata URL   260754
      Object Date           13431
      Object Begin Date     0
      Object End Date       0
      Medium                7215
      Dimensions            75058
```

Credit Line	651
Geography Type	424997
City	452202
State	482335
County	476397
Country	408949
Region	453456
Subregion	462813
Locale	469217
Locus	477438
Excavation	468385
River	482864
Classification	78717
Rights and Reproduction	460427
Link Resource	0
Object Wikidata URL	415802
Metadata Date	484956
Repository	0
Tags	292501
Tags AAT URL	292501
Tags Wikidata URL	292501

dtype: int64

```
[4]: # drop the na values
metobj_df.dropna()
```

```
[4]: Empty DataFrame
Columns: [Object Number, Is Highlight, Is Timeline Work, Is Public Domain,
Object ID, Gallery Number, Department, AccessionYear, Object Name, Title,
Culture, Period, Dynasty, Reign, Portfolio, Constituent ID, Artist Role, Artist
Prefix, Artist Display Name, Artist Display Bio, Artist Suffix, Artist Alpha
Sort, Artist Nationality, Artist Begin Date, Artist End Date, Artist Gender,
Artist ULAN URL, Artist Wikidata URL, Object Date, Object Begin Date, Object End
Date, Medium, Dimensions, Credit Line, Geography Type, City, State, County,
Country, Region, Subregion, Locale, Locus, Excavation, River, Classification,
Rights and Reproduction, Link Resource, Object Wikidata URL, Metadata Date,
Repository, Tags, Tags AAT URL, Tags Wikidata URL]
Index: []

[0 rows x 54 columns]
```

```
[5]: # determine if we can bin by Accession Year
metobj_df["AccessionYear"].unique()
```

```
[5]: array([1979.0, 1980.0, 1967.0, 1916.0, 1909.0, 1964.0, 1970.0, 1904.0,
1976.0, 1938.0, 1918.0, 1946.0, 1960.0, 1910.0, 1922.0, 1923.0,
1930.0, 1934.0, 1936.0, 1937.0, 1939.0, 1941.0, 1947.0, 1948.0,
```

```

1949.0, 1971.0, 1984.0, 1989.0, 1985.0, 1952.0, 1968.0, 1919.0,
nan, 1978.0, 1972.0, 1924.0, 1983.0, 1950.0, 1925.0, 1926.0,
1931.0, 1932.0, 1951.0, 1953.0, 2006.0, 1969.0, 1975.0, 1981.0,
1986.0, 1959.0, 1944.0, 1982.0, 1945.0, 1888.0, 1911.0, 1957.0,
1943.0, 1935.0, 1917.0, 1973.0, 1940.0, 1913.0, 1962.0, 1914.0,
1966.0, 1954.0, 1920.0, 1933.0, 2014.0, 1928.0, 1901.0, 1958.0,
1965.0, 1883.0, 1902.0, 1987.0, 1974.0, 1977.0, 1915.0, 1942.0,
1912.0, 1927.0, 1896.0, 1892.0, 1877.0, 1906.0, 1955.0, 1961.0,
1994.0, 1907.0, 1929.0, 2018.0, 1921.0, 1908.0, 1963.0, 2010.0,
2000.0, 1988.0, 2021.0, 1956.0, 2012.0, 2013.0, 2002.0, 1999.0,
1894.0, 1897.0, 1993.0, 1893.0, 1992.0, 1996.0, 1898.0, 1899.0,
1991.0, 2020.0, 1880.0, 1890.0, 1876.0, 1886.0, 1895.0, 1997.0,
2019.0, 1990.0, 2022.0, 1891.0, 1881.0, 1903.0, 1900.0, 1905.0,
1995.0, 1887.0, 2011.0, 1874.0, 1885.0, 1873.0, 1884.0, 1882.0,
1875.0, 1872.0, 2001.0, 1889.0, 2008.0, 2007.0, 2015.0, 1998.0,
2003.0, 2005.0, 2016.0, 2004.0, 2017.0, 2009.0, 1879.0, 1870.0,
'1933', '1935', '1936', '1937', '1938', '1939', '1940', '1941',
'1942', '1943', '1944', '1945', '1946', '1947', '1948', '1949',
'1950', '1951', '1952', '1953', '1954', '1955', '1956', '1957',
'1958', '1959', '1960', '1961', '1962', '1963', '1964', '1965',
'1966', '1967', '1968', '1969', '1883', '1997', '1996', '1981',
'1978', '1980', '1985', '1983', '1979', '1991', '1993', '1990',
'1994', '2014', '2020', '2015', '2021', '2018', '2019', '2017',
'2022', '1995', '2000', '1984', '1998', '1918', '1972', '1999',
'1975', '1989', '1987', '1973', '1974', '1982', '2001', '2003',
'2010', '2013', '2002', '1992', '1976', '2004', '2005', '1970',
'1977', '2005-02-15', '2006', '2011', '2007', '2008', '2009',
1878.0, 1871.0, '1920', '1921', '1922', '1923', '1924', '1925',
'1926', '1927', '1928', '1929', '1930', '1931', '1932', '1934',
'2016', '1986', '1988', '2012', '1971', '1909', '1889', '1906',
'1894', '1907', '1908', '1910', '1911', '1912', '1915', '1913',
'1914', '2022-05-20', '1916', '1917', '1919', '1904', '1898',
'1905', '1899', '1874', '1890', '1886', '1901', '1902', '1903',
'1896', '1900', 2023.0, '2023', '1891', '1893', '2022-02-09',
'1953-05-28', '1956-08-24', '1885', '1978-12-12', '1977-08-08',
'1980-06-18', '1978-03-07', '1963-06-17', '1959-10-16',
'1960-11-15', '1960-06-25', '1961-01-10', '1968-12-12',
'1966-04-29', '1967-08-12', '1967-12-04', '1969-07-10',
'1969-07-17', '1970-09-08', '1950-07-21', '1953-01-19',
'1955-05-24', '1962-10-15', '1959-12-28'], dtype=object)

```

```

[6]: # Values aren't clean so we have to manipulate the string to clean them
# strip whitespace
metobj_df.columns = metobj_df.columns.str.strip()

```

```

[7]: # Convert to string
col = metobj_df["AccessionYear"].astype(str).str.strip()

```

```

# extract the year from each value
years = col.str.extract(r"(\d{4})")[0]

# ensure the years are numeric
metobj_df["AccessionYear_clean"] = pd.to_numeric(years, errors="coerce")

# injection due to noticing a stray "." after my values & it bugged me
metobj_df["AccessionYear_clean"] = (
    metobj_df["AccessionYear_clean"].astype("Int64")
)

# no need to bin if I did the manipulate strings part of Ch7

```

```

[8]: # relook at Accession Year
metobj_df["AccessionYear_clean"].unique()

```

```

[8]: <IntegerArray>
[1979, 1980, 1967, 1916, 1909, 1964, 1970, 1904, 1976, 1938,
...
2005, 2016, 2004, 2017, 2009, 1879, 1870, 1878, 1871, 2023]
Length: 155, dtype: Int64

```

Ch 10 transformations

- Grouping with Dicts/Series
- Cross Tab

```

[9]: # Determine departments
metobj_df["Department"].unique()

```

```

[9]: array(['The American Wing', 'European Sculpture and Decorative Arts',
'Modern and Contemporary Art', 'Arms and Armor', 'Medieval Art',
'Asian Art', 'Islamic Art', 'Costume Institute',
'Arts of Africa, Oceania, and the Americas', 'Drawings and Prints',
'Greek and Roman Art', 'Photographs', 'Ancient Near Eastern Art',
'Egyptian Art', 'European Paintings', 'Robert Lehman Collection',
'The Cloisters', 'Musical Instruments', 'The Libraries'],
dtype=object)

```

```

[10]: # Create the groups
Dept_Groups = {
    "The American Wing": "Western Art",
    "European Sculpture and Decorative Arts": "Western Art",
    "European Paintings": "Western Art",
    "Greek and Roman Art": "Ancient Western",
    "Ancient Near Eastern Art": "Ancient",
    "Egyptian Art": "Ancient",

```



```

    "Medieval Art": "Historical",
    "Arms and Armor": "Historical",
    "Asian Art": "Non-Western",
    "Islamic Art": "Non-Western",
    "Arts of Africa, Oceania, and the Americas": "Non-Western",
    "Modern and Contemporary Art": "Modern",
    "Drawings and Prints": "Works on Paper",
    "Photographs": "Works on Paper",
    "Costume Institute": "Special Collections",
    "Musical Instruments": "Special Collections",
    "The Libraries": "Special Collections",
    "Robert Lehman Collection": "Special Collections",
    "The Cloisters": "Historical"
}

```

```

[11]: # apply the grouping
metobj_df["Dept_Group"] = metobj_df["Department"].map(Dept_Groups)

```

```

[12]: # create a cross tab
metobj_df["Dept_Group"].value_counts()

```

```

[12]: Dept_Group
Works on Paper      210089
Non-Western         64940
Western Art         64209
Special Collections  39999
Ancient             34192
Ancient Western     33726
Historical          23105
Modern              14696
Name: count, dtype: int64

```

```

[13]: # preview
metobj_df.head()

```

```

[13]:  Object Number  Is Highlight  Is Timeline Work  Is Public Domain  Object ID  \
0      1979.486.1         False          False          False           1
1      1980.264.5         False          False          False           2
2         67.265.9         False          False          False           3
3         67.265.10        False          False          False           4
4         67.265.11        False          False          False           5

   Gallery Number      Department AccessionYear Object Name  \
0              NaN  The American Wing         1979.0      Coin
1              NaN  The American Wing         1980.0      Coin
2              NaN  The American Wing         1967.0      Coin
3              NaN  The American Wing         1967.0      Coin

```

4	NaN	The American Wing	1967.0	Coin
---	-----	-------------------	--------	------

	Title	...	Rights and Reproduction	\
0	One-dollar Liberty Head Coin	...	NaN	
1	Ten-dollar Liberty Head Coin	...	NaN	
2	Two-and-a-Half Dollar Coin	...	NaN	
3	Two-and-a-Half Dollar Coin	...	NaN	
4	Two-and-a-Half Dollar Coin	...	NaN	

	Link Resource	Object Wikidata	URL	\
0	http://www.metmuseum.org/art/collection/search/1		NaN	
1	http://www.metmuseum.org/art/collection/search/2		NaN	
2	http://www.metmuseum.org/art/collection/search/3		NaN	
3	http://www.metmuseum.org/art/collection/search/4		NaN	
4	http://www.metmuseum.org/art/collection/search/5		NaN	

	Metadata Date	Repository	Tags	Tags	AAT	URL	\
0	NaN	Metropolitan Museum of Art, New York, NY	NaN			NaN	
1	NaN	Metropolitan Museum of Art, New York, NY	NaN			NaN	
2	NaN	Metropolitan Museum of Art, New York, NY	NaN			NaN	
3	NaN	Metropolitan Museum of Art, New York, NY	NaN			NaN	
4	NaN	Metropolitan Museum of Art, New York, NY	NaN			NaN	

	Tags Wikidata	URL	AccessionYear_clean	Dept_Group
0	NaN		1979	Western Art
1	NaN		1980	Western Art
2	NaN		1967	Western Art
3	NaN		1967	Western Art
4	NaN		1967	Western Art

[5 rows x 56 columns]

You may be wondering why chapter 10's transformations & cleanings are before chapter 8; it's because I went through and did ones I had recently performed, and then went back and did other actions. However, a little fun fact I forgot about, when you restart your kernel and then rerun the blocks in order, an index creation will 100% cause an error when you try to group on that column later. So I rearranged them. In hindsight, I could have duplicated the column and grouped it off of that one, though. I'll have to keep that in mind for the future.

Ch 8 transformations

- Create Hierarchical Index
- Pivot the Data

```
[14]: # Set the index
metobj_df = metobj_df.set_index(["Department", "Object Number"])
```

```
[15]: # Pivot the data
pivot = metobj_df.reset_index().pivot_table(
    index="Department",
    values="Object Number",
    aggfunc="count"
)
```

```
[16]: # preview
metobj_df.head()
```

```
[16]:
```

Department	Object Number	Is Highlight	Is Timeline Work	\
The American Wing	1979.486.1	False	False	
	1980.264.5	False	False	
	67.265.9	False	False	
	67.265.10	False	False	
	67.265.11	False	False	

Department	Object Number	Is Public Domain	Object ID	Gallery Number	\
The American Wing	1979.486.1	False	1	NaN	
	1980.264.5	False	2	NaN	
	67.265.9	False	3	NaN	
	67.265.10	False	4	NaN	
	67.265.11	False	5	NaN	

Department	Object Number	AccessionYear	Object Name	\
The American Wing	1979.486.1	1979.0	Coin	
	1980.264.5	1980.0	Coin	
	67.265.9	1967.0	Coin	
	67.265.10	1967.0	Coin	
	67.265.11	1967.0	Coin	

Department	Object Number	Title	Culture Period	\
The American Wing	1979.486.1	One-dollar Liberty Head Coin	NaN	NaN
	1980.264.5	Ten-dollar Liberty Head Coin	NaN	NaN
	67.265.9	Two-and-a-Half Dollar Coin	NaN	NaN
	67.265.10	Two-and-a-Half Dollar Coin	NaN	NaN
	67.265.11	Two-and-a-Half Dollar Coin	NaN	NaN

Department	Object Number	... Rights and Reproduction	\
The American Wing	1979.486.1	NaN	
	1980.264.5	NaN	
	67.265.9	NaN	

67.265.10	...	NaN
67.265.11	...	NaN

Link

Resource \	Department	Object Number
	The American Wing	1979.486.1
		1980.264.5
		67.265.9
		67.265.10
		67.265.11

<http://www.metmuseum.org/art/collection/search/1>
<http://www.metmuseum.org/art/collection/search/2>
<http://www.metmuseum.org/art/collection/search/3>
<http://www.metmuseum.org/art/collection/search/4>
<http://www.metmuseum.org/art/collection/search/5>

Department	Object Number	Object Wikidata URL	Metadata Date \
The American Wing	1979.486.1	NaN	NaN
	1980.264.5	NaN	NaN
	67.265.9	NaN	NaN
	67.265.10	NaN	NaN
	67.265.11	NaN	NaN

Department	Object Number	Repository \
The American Wing	1979.486.1	Metropolitan Museum of Art, New York, NY
	1980.264.5	Metropolitan Museum of Art, New York, NY
	67.265.9	Metropolitan Museum of Art, New York, NY
	67.265.10	Metropolitan Museum of Art, New York, NY
	67.265.11	Metropolitan Museum of Art, New York, NY

Department	Object Number	Tags	Tags AAT	URL	Tags Wikidata	URL \
The American Wing	1979.486.1	NaN	NaN		NaN	
	1980.264.5	NaN	NaN		NaN	
	67.265.9	NaN	NaN		NaN	
	67.265.10	NaN	NaN		NaN	
	67.265.11	NaN	NaN		NaN	

Department	Object Number	AccessionYear_clean	Dept_Group
The American Wing	1979.486.1	1979	Western Art
	1980.264.5	1980	Western Art
	67.265.9	1967	Western Art
	67.265.10	1967	Western Art

[5 rows x 54 columns]

Ch 11 transformations

- Convert between string and datetime
- Generate a date range

```
[17]: # convert accession yr to datetime
metobj_df["AccessionYear_dt"] = pd.to_datetime(
    metobj_df["AccessionYear"],
    errors="coerce"
)
```

```
[18]: # Generate a date range
date_range = pd.date_range(start="1800-01-01", periods=100, freq="Y")
```

```
[19]: # preview
date_range
```

```
[19]: DatetimeIndex(['1800-12-31', '1801-12-31', '1802-12-31', '1803-12-31',
                    '1804-12-31', '1805-12-31', '1806-12-31', '1807-12-31',
                    '1808-12-31', '1809-12-31', '1810-12-31', '1811-12-31',
                    '1812-12-31', '1813-12-31', '1814-12-31', '1815-12-31',
                    '1816-12-31', '1817-12-31', '1818-12-31', '1819-12-31',
                    '1820-12-31', '1821-12-31', '1822-12-31', '1823-12-31',
                    '1824-12-31', '1825-12-31', '1826-12-31', '1827-12-31',
                    '1828-12-31', '1829-12-31', '1830-12-31', '1831-12-31',
                    '1832-12-31', '1833-12-31', '1834-12-31', '1835-12-31',
                    '1836-12-31', '1837-12-31', '1838-12-31', '1839-12-31',
                    '1840-12-31', '1841-12-31', '1842-12-31', '1843-12-31',
                    '1844-12-31', '1845-12-31', '1846-12-31', '1847-12-31',
                    '1848-12-31', '1849-12-31', '1850-12-31', '1851-12-31',
                    '1852-12-31', '1853-12-31', '1854-12-31', '1855-12-31',
                    '1856-12-31', '1857-12-31', '1858-12-31', '1859-12-31',
                    '1860-12-31', '1861-12-31', '1862-12-31', '1863-12-31',
                    '1864-12-31', '1865-12-31', '1866-12-31', '1867-12-31',
                    '1868-12-31', '1869-12-31', '1870-12-31', '1871-12-31',
                    '1872-12-31', '1873-12-31', '1874-12-31', '1875-12-31',
                    '1876-12-31', '1877-12-31', '1878-12-31', '1879-12-31',
                    '1880-12-31', '1881-12-31', '1882-12-31', '1883-12-31',
                    '1884-12-31', '1885-12-31', '1886-12-31', '1887-12-31',
                    '1888-12-31', '1889-12-31', '1890-12-31', '1891-12-31',
                    '1892-12-31', '1893-12-31', '1894-12-31', '1895-12-31',
                    '1896-12-31', '1897-12-31', '1898-12-31', '1899-12-31'],
                    dtype='datetime64[ns]', freq='A-DEC')
```

[20]: *# That didn't tell me a whole lot so let's try a different method that I was*
↳ originally going to partially do for chapter 7.

```
# bin 'em
bins = list(range(1850, 2051, 50))

# label 'em
labels = [
    "1850-1899",
    "1900-1949",
    "1950-1999",
    "2000-2049"
]

#
metobj_df["YearGroup_50"] = pd.cut(
    metobj_df["AccessionYear_clean"],
    bins=bins,
    labels=labels,
    right=False
)

# count 'em
metobj_df["YearGroup_50"].value_counts().sort_index()
```

[20]: YearGroup_50
1850-1899 25223
1900-1949 180376
1950-1999 202224
2000-2049 73271
Name: count, dtype: int64